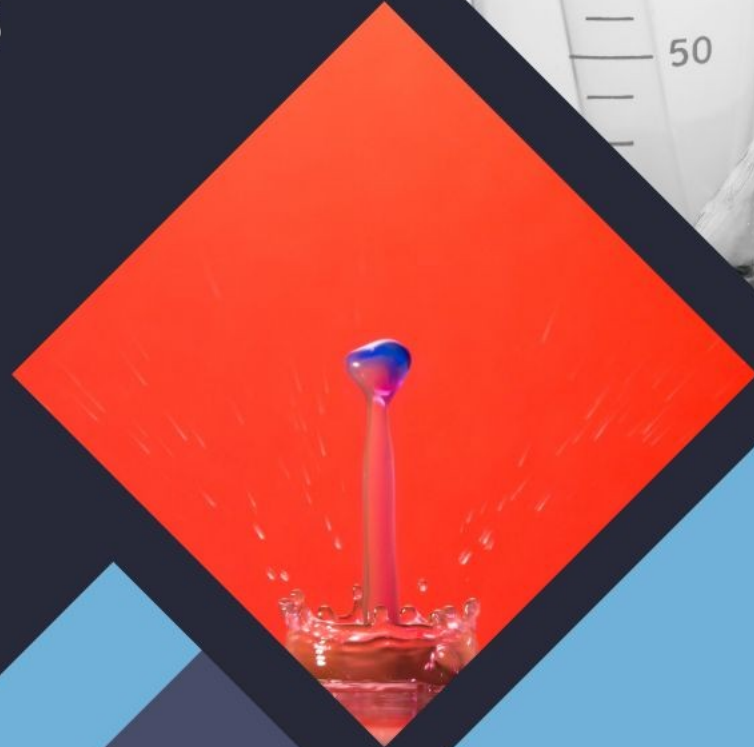


Exploring Selected Gravimetric Methods and Their Practical Applications

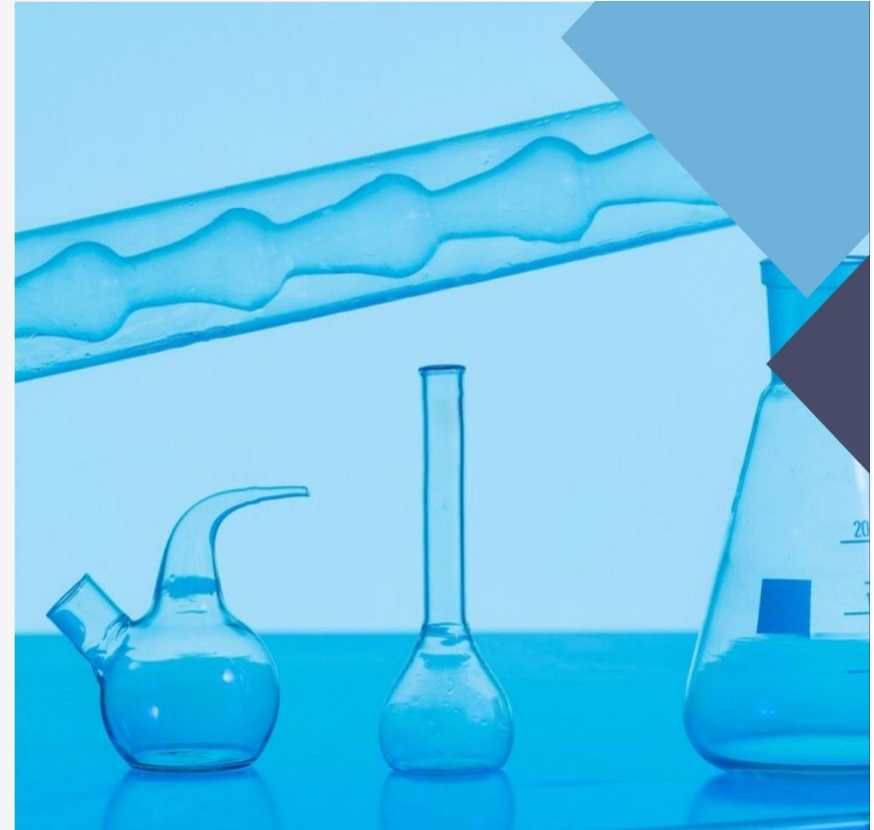


OBJECTIVE

- ❖ **DEFINE GRAVIMETERY.**
- ❖ **IDENTITY TYPES OF
SELECTIVE
GRAVIMETRIC METHODS**
- ❖ **UNDERSTANDING THEIR
APPLICATION.**

Introduction to Gravimetric Analysis

- Gravimetry encompasses all techniques in which we measure mass or a change in mass.
- It is an analytical technique where mass or changes in mass are measured to determine the composition or concentration of a substance.
- It is the oldest and most fundamental method of measurement in analytical chemistry
- Known for its precision and reliability in quantifying substances, gravimetry provides highly accurate results, especially for pure substances.
- Widely used in chemistry, pharmaceuticals, environmental testing, and materials science for determining concentrations, purities, and compositions.

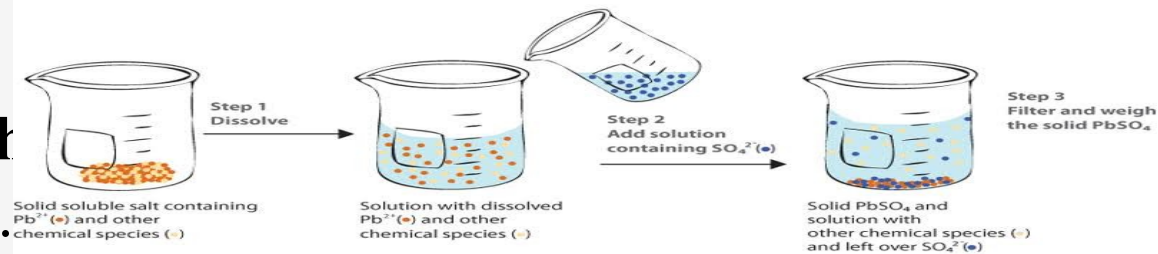


Types of Gravimetric Methods

- There are 4 major types of gravimetric methods of analysis:
- **Precipitation Gravimetry:** The analyte forms an insoluble precipitate, which is isolated, purified, and weighed to determine its quantity
- **Volatilization Gravimetry:** The analyte is heated to a volatile state, and the remaining residue is weighed.
- **Particulate Gravimetry:** Particles are separated from a sample (e.g., air or water) through filtration or settling, then dried and weighed to quantify particulate matter.
- **Electrogravimetry:** The analyte is deposited electrochemically on an electrode, and its mass is measured.
- Other types of gravimetric methods include thermogravimetry and crystallization gravimetry.

Precipitation Gravimetry

- Precipitation gravimetry involves adding a precipitating reagent to a solution containing the analyte, resulting in the formation of an insoluble compound (precipitate).
- Typically, the precipitate forms through a simple double decomposition reaction (metathesis) reaction between the analyte and the precipitant, though any reaction generating a precipitate can be used.
- Key Attributes of this reaction includes
 1. The precipitate must have **low solubility, high composition** for accurate mass determination.
 2. The precipitate must be easily **separated** from the reaction mixture to ensure accurate results.



Process precipitation gravimetry

- **Gravimetric Analysis Process:**

In precipitation gravimetric analysis, the analyte reacts with a specific reagent to form an insoluble precipitate. The precipitate is then separated from the solution by filtration and washed with a solvent to remove any soluble impurities. Afterward, the precipitate is dried or heated to eliminate moisture or volatile substances.

- **Weighing and Calculation:**

Once the precipitate is fully dried, it is carefully weighed using a precise analytical balance. The mass of the dried precipitate is used to calculate the amount of analyte in the sample, providing an accurate and reliable result.

Application

Quantitative Applications

- **Quantitative Analysis:** Measures the amount of analyte in a sample by calculating the mass of the precipitate formed during the reaction.
- **Inorganic Analysis:** To determine the concentration of inorganic ions and elements, such as metals (e.g., copper, iron) in ores, alloys, and environmental samples.
- **Organic Analysis:** To quantify organic compounds by precipitating substances like halides, sulfates, or phosphates,

Qualitative Applications

- **Ion Identification:** identifying specific ions in a sample by observing the formation of distinct precipitates upon the addition of reagents.
- **Spot Testing:** Employed for rapid identification of analytes, particularly in field analysis or quick qualitative assessments.
- **Qualitative Analysis of Organic Compounds:** identify the presence of organic analytes by forming a precipitate when reacting with specific reagents.

Volatilization Gravimetry

- Volatilization Gravimetry is a quantitative analytical technique used to determine the amount of a substance in a sample by measuring the mass of the substance that volatilizes (evaporates) under controlled conditions
- It involves heating a sample to convert specific components of the sample into a gas or vapor phase (volatilize), leaving behind a solid residue. The remaining solid residue, after the volatile components have been removed, is then weighed.
- The mass of the residue provides the necessary data for determining the analyte's concentration or the composition of the sample.
- The key essence of this technique lies in the **mass loss** that occurs during heating. The weight difference between the initial and final states of the sample reflects the amount of volatile substance that has been removed

Process of Volatilization Gravimetry

- **Volatilization Process**

The sample is first prepared by grinding or drying to ensure consistency and facilitate the volatilization of specific components. Its initial mass is then measured precisely using a balance, providing a reference for mass loss during the process. The sample is heated under controlled conditions, allowing only the volatile components to evaporate while preventing decomposition.

- **Post-Volatilization Analysis:**

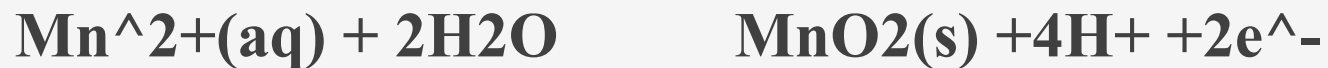
After volatilization, the remaining solid residue is cooled and reweighed to determine its mass. The difference between the initial and final weights represents the mass of the volatile components that were lost. Data is analyzed to quantify the volatile substances and assess the sample's composition.

Application of Volatilization Gravimetry

- **Material Characterization:** Assessing thermal stability and decomposition of materials like polymers and metals.
- **Pharmaceuticals:** Evaluating drug stability and moisture content in formulations.
- **Environmental Studies:** Analyzing waste composition and thermal degradation of pollutants.
- **Food Science:** Investigating thermal properties and moisture levels in food products.
- **Polymer Science:** Studying thermal degradation and stability of polymers and additives.
- **Nanotechnology:** Characterizing the thermal behavior of nanomaterials and composites.
- **Metallurgy:** Analyzing thermal properties of metal alloys and powder

Electrogravimetry

- Electrogravimetry is a method in which the analyte is quantitatively determined by its electrochemical deposition onto an electrode analyte (either the cathode or the anode) in electrochemical cell.
- It involves applying an electric current to a solution containing the analyte, causing the analyte to deposit as a solid onto a conductive surface. The weight of this deposit is directly related to the amount of analyte present in the sample
- Electrogravimetry combines both **gravimetric** and **electrochemical** principles.
- It is governed by **Faraday's laws of electrolysis**, which relate the quantity of electricity used in the electrolysis process to the mass of the material deposited.
- The oxidation of Mn and deposition as MnO₂ on Pt anode is one example of electrogravimetry.



Reduction also may be use in electrogravimetry.



Zinc ion reduced and deposited as zinc metal on Pt cathode.



Process of Electrogravimetry

•Electrolytic Cell Setup:

The sample, which contains metal ions, is dissolved in an electrolyte that allows the flow of current and facilitates the electrochemical deposition of the analyte. The electrolyte must support the deposition process without interfering with the reaction.

•Electrolytic Deposition and Cooling:

A constant current or voltage is applied, causing metal ions to be reduced and deposited as a solid at the working electrode. After deposition, the electrode is removed, washed, and dried to eliminate any excess solvent or impurities.

•Weighing and Calculation:

The electrode is weighed both before and after deposition. The increase in mass corresponds to the amount of analyte deposited. Using Faraday's Law, the mass of the deposit is calculated, and the results are analyzed by comparing the calculated mass to standards or calibration curves.

Application of Electrogravimetry

- **Quantitative Analysis** : Used to determine the concentration of specific metal ions in a sample by measuring the mass of the metal deposited.
- **Separation** : It can help separate components in a mixture by selectively depositing a desired metal while leaving others in solution.
- **Electrosynthesis** : Electrogravimetry is utilized in electrosynthesis processes to produce pure metals or metal alloys by electrochemical deposition.
- **Purification Process** : It plays a key role in purifying metals by selectively removing impurities through controlled electrolysis, leading to high-purity metal deposition.

Particulate Gravimetry

- Particulate gravimetry is a type of gravimetric analysis that involves the measurement of the mass of particulate matter in a sample.
- This method is commonly used in environmental monitoring to determine the concentration of airborne particles, such as dust, soot, and other pollutants.
- In particulate gravimetry, a sample of air is collected on a filter, and the filter is then weighed before and after the collection to determine the mass of the particulate matter.

Cont.....

- This method provides a direct measurement of the amount of particulate matter present in the air, and it is widely used in air quality monitoring and pollution control efforts.
- Particulate gravimetry is also used in industrial settings to monitor dust levels in workplaces and to ensure compliance with occupational health and safety regulations.
- Overall, particulate gravimetry is an important tool for assessing air quality and controlling particulate pollution in various environments.

Some advantages and limitations of gravimetric methods

Advantages

- High Accuracy and Precision
- Simplicity
- Cost-Effectiveness
- Independence from Calibration
- Robust against Interferences

Limitations

- Time-Consuming
- Sample Condition
- Requires Purity
- Non-selective
- Limited Detection Range

Some key pharmaceutical industrial application of Gravimetry

- **Mass Measurement of Substances** : measure the mass of active pharmaceutical ingredients (APIs) and excipients, ensuring precise formulation of medications.
- **Purity Determination** :determining the purity of substances by measuring mass changes during the precipitation of compounds
- **Content Uniformity Testing** : assess the uniformity of content in dosage forms, such as tablets and capsule.
- **Dissolution Testing** : used to measure the amount of drug released from a dosage form over time, providing insights into the dissolution characteristics of the formulation
- **Quality control** : quality assurance processes to verify that products meet specified standards and regulatory requirements

Summary

- Selective gravimetric methods are valuable in quantitative analysis because they allow for the specific determination of individual components in a complex sample. These methods are widely used in analytical chemistry, environmental monitoring, and various industries where precise measurement of specific substances is required.
- Overall, gravimetry, and selective gravimetric methods such as volatilization gravimetry, play an important role in quantitative analysis and are widely used in various industries and research fields.

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THANK YOU!

Group Mememebers

- **Melekete Yohannes** **Ugr/2036/15**
- **Mengesha Abiyu** **Ugr/1224/15**
- **Metusal Zeleke** **Ugr/0754/15**
- **Misrak Eyasu** **Ugr/6146/15**
- **Nahil Getu** **Ugr/0453/15**
- **Nahom Asnake** **Ugr/5634/15**
- **Nahom Solomon** **Ugr/9810/15**
- **Niya Gedefa** **Ugr/7896/15**
- **Pawlos Hailu** **Ugr/0515/15**
- **Rediet Yismaw** **Ugr/0526/15**